

Breast Cancer Treatment (PDQ) - Breast Cancer Knowledge Base

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Subtype: adult female breast cancer

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Breast Cancer Treatment (Adult) (PDQ®) — Health Professional Version

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of breast cancer. It is intended as a resource to inform and assist clinicians who care for cancer patients. It does not provide formal guidelines or recommendations for making health care decisions.

This summary is reviewed regularly and updated as necessary by the PDQ Adult Treatment Editorial Board, which is editorially independent of the National Cancer Institute (NCI). This summary discusses primary epithelial breast cancers in women. Related PDQ summaries cover Male Breast Cancer Treatment and Breast Cancer Treatment During Pregnancy.

General Information About Breast Cancer

Incidence and Mortality

Estimated new cases and deaths from breast cancer (women only) in the United States in 2019:

- New cases: 268,600
- Deaths: 41,760

Breast cancer is the most common noncutaneous cancer in U.S. women, with an estimated 62,930 cases of in situ disease and 268,600 cases of invasive disease in 2019. Thus, fewer than one of six women diagnosed with breast cancer die of the disease. Men account for 1% of breast cancer cases and deaths.

Widespread adoption of screening increases breast cancer incidence in a given population and changes the characteristics of cancers detected, with increased incidence of lower-risk cancers, premalignant lesions, and ductal carcinoma in situ (DCIS).

Anatomy

The female breast contains the nipple, areola, lymph nodes, lobes, lobules, and ducts, surrounded by fatty tissue, chest wall, ribs, and muscle.

Risk Factors

- Family health history

- Major inheritance susceptibility, including germline mutation of the *BRCA1* and *BRCA2* genes and other breast cancer susceptibility genes
- Alcohol intake
- Breast tissue density (mammographic)
- Estrogen (endogenous)
- Menstrual history (early menarche/late menopause)
- Nulliparity
- Older age at first birth
- Hormone therapy history (combination estrogen plus progestin hormone replacement therapy)
- Obesity (postmenopausal)
- Personal history of breast cancer
- Personal history of benign breast disease (proliferative forms)
- Radiation exposure to breast/chest

Of all women with breast cancer, 5–10% may have a germline mutation of *BRCA1* and *BRCA2*. Specific mutations are more common in women of Jewish ancestry. Estimated lifetime risk of developing breast cancer for *BRCA1/BRCA2* mutation carriers is 40–85%. Carriers with a history of breast cancer have an increased risk of contralateral disease that may be as high as 5% per year. Male *BRCA2* mutation carriers also have increased breast cancer risk. Mutations in either gene also confer increased risk of ovarian cancer and other primary cancers.

Protective Factors

- Estrogen use (after hysterectomy)
- Exercise
- Early pregnancy
- Breast feeding
- Selective estrogen receptor modulators (SERMs)
- Aromatase inhibitors or inactivators
- Risk-reducing mastectomy
- Risk-reducing oophorectomy or ovarian ablation

Screening

Clinical trials have established that screening asymptomatic women using mammography, with or without clinical breast examination, decreases breast cancer mortality.

Diagnosis

Patient evaluation generally includes: confirmation of the diagnosis, evaluation of the stage of disease, and selection of therapy.

Tests and procedures used to diagnose breast cancer:

- Mammography
- Ultrasound
- Breast MRI, if clinically indicated

- Biopsy

Contralateral disease: Breast cancer can be multicentric and bilateral. Bilateral disease is somewhat more common with infiltrating lobular carcinoma. At 10 years after diagnosis, risk of a primary breast cancer in the contralateral breast ranges from 3–10%, though endocrine therapy decreases that risk. When BRCA1/BRCA2 mutation carriers were diagnosed before age 40, the risk of contralateral breast cancer reached nearly 50% in the ensuing 25 years.

Prognostic and Predictive Factors

Prognosis and therapy selection may be influenced by:

- Menopausal status
- Stage of disease
- Grade of the primary tumor
- Estrogen receptor (ER) and progesterone receptor (PR) status
- HER2/neu overexpression and/or amplification
- Histologic type (favorable types: mucinous, medullary, tubular carcinomas)

Classification based on ER, PR, and HER2/neu:

- Hormone receptor positive
- HER2/neu positive
- Triple negative (ER, PR, and HER2/neu negative)

Gene profile tests:

- **MammaPrint:** 70-gene signature classifying tumors into high- and low-risk prognostic categories. FDA-approved.
- **Oncotype DX:** 21-gene recurrence score, applies to hormone receptor–positive breast cancer.
- Recurrence score <18: low risk
- Recurrence score ≥ 18 and <31: intermediate risk
- Recurrence score ≥ 31 : high risk

Key evidence — Oncotype DX:

- NSABP B-14 trial: 10-year distant recurrence risk with tamoxifen was 7% (low score), 14% (intermediate score), 31% (high score).
- **TAILORx trial** (6,907 women, recurrence score 11–25, randomized to endocrine therapy alone vs. plus chemotherapy): 9-year invasive DFS was 83.3% (endocrine alone) vs. 84.3% (endocrine + chemo) — met noninferiority criterion. Chemotherapy may benefit women younger than 50 with recurrence scores 16–25 depending on clinical risk.
- Low-risk score (<11) patients had very low 5-year recurrence rates with endocrine therapy alone: 93.8% invasive DFS at 5 years, 98.0% OS at 5 years.

Key evidence — MammaPrint (MINDACT trial): Women with high clinical risk but low genomic risk who skipped chemotherapy had 94.7% 5-year survival without distant metastases (met noninferiority endpoint).

Although BRCA1/BRCA2 mutations predispose to breast cancer, prognostic data for carriers who develop breast cancer are conflicting; these women are at greater risk of contralateral breast cancer.

Histopathologic Classification of Breast Cancer

Infiltrating/invasive ductal carcinoma is the most common histologic type, comprising 70–80% of all cases.

Tumor Location	Histologic Subtype
Ductal	Intraductal (in situ); Invasive NOS; Comedo; Inflammatory; Medullary with lymphocytic infiltrate; Mucinous (colloid); Papillary; Scirrhous; Tubular; Other
Lobular	Invasive with predominant in situ component; Invasive
Nipple	Paget disease NOS; Paget disease with intraductal carcinoma; Paget disease with invasive ductal carcinoma
Other	Undifferentiated carcinoma; Metaplastic

Tumor subtypes that occur in the breast but are not considered typical breast cancers: Phyllodes tumor, Angiosarcoma, Primary lymphoma.

Stage Information for Breast Cancer

The AJCC staging system groups patients by prognosis, incorporating tumor size, lymph node status, ER/PR levels, HER2/neu status, tumor grade, menopausal status, and general health.

Biomarker Status Standards

- **ER expression:** measured by immunohistochemistry (IHC). Any staining $\geq 1\%$ of cells = positive.
- **PR expression:** measured by IHC. Any staining $\geq 1\%$ of cells = positive.
- **HER2 expression:** measured by IHC (protein expression) or in situ hybridization/ISH (gene copy number).
- IHC: Negative (0/1+), Equivocal (2+), Positive (3+)
- ISH (dual probe): Negative (ratio < 2.0 AND copy number < 4); Equivocal (ratio < 2.0 AND copy number ≥ 4 but < 6); Positive (ratio ≥ 2.0 , or copy number ≥ 6 regardless of ratio)
- ISH (single probe): Negative (< 4 copies); Equivocal (≥ 4 but < 6 copies); Positive (≥ 6 copies)

TNM Staging (AJCC 8th edition)

T (Primary Tumor):

Category	Criteria
TX	Cannot be assessed
T0	No evidence of primary tumor
Tis	DCIS
Tis (Paget)	Paget disease of the nipple without invasive carcinoma/DCIS
T1 (mi/a/b/c)	$\leq 20\text{mm}$, subdivided by size ($\leq 1\text{mm}$ / 1–5mm / 5–10mm / 10–20mm)
T2	$> 20\text{mm}$ but $\leq 50\text{mm}$
T3	$> 50\text{mm}$

Category	Criteria
T4 (a/b/c/d)	Any size with chest wall/skin extension; T4d = inflammatory carcinoma

N (Clinical, cN):

Category	Criteria
cNX	Cannot be assessed
cN0	No regional lymph node metastases
cN1	Movable ipsilateral Level I, II axillary node(s)
cN2	Fixed/matted ipsilateral Level I, II axillary nodes, or internal mammary nodes without axillary involvement
cN3	Infraclavicular, internal mammary + axillary, or supraclavicular node involvement

N (Pathological, pN):

Category	Criteria
pNX	Cannot be assessed
pN0	No regional lymph node metastasis (or isolated tumor cells only)
pN1	Micrometastases or 1–3 axillary nodes; and/or internal mammary micro/macrometastases by sentinel biopsy
pN2	4–9 axillary nodes, or clinically detected internal mammary nodes without axillary involvement
pN3	≥10 axillary nodes, infraclavicular nodes, or supraclavicular nodes; or combinations of internal mammary + axillary involvement

M (Distant Metastasis):

Category	Criteria
M0	No clinical/radiographic evidence of distant metastases
cM0(i+)	No clinical evidence, but tumor cells ≤0.2mm detected microscopically in blood/marrow/nonregional nodes
cM1	Distant metastases detected clinically/radiographically
pM1	Histologically proven distant metastases

Histologic Grade (G)

- G1: Low grade (favorable), SBR score 3–5
- G2: Intermediate grade, SBR score 6–7

- G3: High grade (unfavorable), SBR score 8–9

Stage Groups (Anatomic — simplified)

Stage	TNM
0	Tis, N0, M0
IA	T1, N0, M0
IB	T0–T1, N1mi, M0
IIA	T0–T1 with N1, M0; or T2, N0, M0
IIB	T2, N1, M0; or T3, N0, M0
IIIA	T0–T2 with N2, M0; or T3 with N1–N2, M0
IIIB	T4 with N0–N2, M0
IIIC	Any T, N3, M0
IV	Any T, Any N, M1

Note: The full Clinical and Pathological Prognostic Stage Groups incorporate grade, HER2, ER, and PR status alongside TNM — for example, a T1, N0, M0 tumor can range from Stage IA (grade 1, HR-positive) to Stage IB (grade 3, HR-negative) depending on biomarker profile. These detailed lookup tables are required for staging in the United States and should be consulted directly from the source for precise stage assignment given a full biomarker profile.

Staging following neoadjuvant therapy is denoted with a "yc" or "yp" prefix. There is no anatomic stage group assigned if there is a complete pathological response (pCR) to neoadjuvant therapy (e.g., ypT0, ypN0, cM0).

[Note: This is a condensed, cleaned excerpt of the Breast Cancer Treatment PDQ summary covering General Information, Histopathologic Classification, and Staging. The full source document continues with detailed treatment options by stage (DCIS, Stage I–III, Inflammatory Breast Cancer, Stage IV/Metastatic/Recurrent), including surgery, radiation, endocrine therapy, chemotherapy, HER2-targeted therapy, and CDK4/6 inhibitor evidence. Fetch the remaining sections from the source URL to complete the corpus.]